

**SECONDARY 3 EXPRESS — END-OF-YEAR EXAMINATION**  
**ADDITIONAL MATHEMATICS 4049**  
**WORKED SOLUTIONS & MARKING SCHEME**

Question • Working • Marks & examiner's notes

**Mark codes:**

- M** method mark — awarded for a correct method, even if the arithmetic that follows is wrong
- A** accuracy mark — awarded for a correct answer or correct intermediate value; dependent on the preceding M mark
- B** independent mark — awarded for a correct statement, value or result on its own, with no method required

*Total: 90 marks. Follow-through (ft) applies throughout: a correct method applied to a candidate's own earlier incorrect value still earns the M mark.*

**Question 1** — Surds — area of a triangle**[3]**

The triangle  $DEF$  has an area of  $\frac{1}{4}(7\sqrt{3}+9)$  cm<sup>2</sup>. The length of  $DE$  is  $(\sqrt{3}+2)$  cm and angle  $EDF$  is  $60^\circ$ . Find, without using a calculator, the length of  $DF$ , in cm, in the form  $(a+b\sqrt{3})$ .

$$\text{Area} = \frac{1}{2}(DE)(DF)\sin \angle EDF, \text{ so } \frac{1}{2}(\sqrt{3}+2)(DF)\left(\frac{\sqrt{3}}{2}\right) = \frac{1}{4}(7\sqrt{3}+9) \quad \text{M1}$$

$$\sqrt{3}(\sqrt{3}+2)(DF) = 7\sqrt{3}+9 \Rightarrow (3+2\sqrt{3})(DF) = 7\sqrt{3}+9$$

$$DF = \frac{7\sqrt{3}+9}{3+2\sqrt{3}} \times \frac{3-2\sqrt{3}}{3-2\sqrt{3}} = \frac{3\sqrt{3}-15}{-3} \quad \text{M1}$$

$$\text{Answer: } DF = 5 - \sqrt{3} \text{ cm, so } a = 5 \text{ and } b = -1. \quad \text{A1}$$

**Note:** The  $\frac{1}{2}$  and the  $\sin 60^\circ = \frac{\sqrt{3}}{2}$  must both appear for the first M1. Denominator

$(3)^2 - (2\sqrt{3})^2 = 9 - 12 = -3$  is negative — candidates who drop the sign get  $\sqrt{3}-5$  and lose the A1.

**Question 2** — Surds — rationalising the denominator**[4]**

(a) Simplify  $\frac{7}{3-\sqrt{2}}$  in the form  $a+b\sqrt{2}$ . [2]

$$\frac{7}{3-\sqrt{2}} \times \frac{3+\sqrt{2}}{3+\sqrt{2}} = \frac{7(3+\sqrt{2})}{9-2} = \frac{7(3+\sqrt{2})}{7} \quad \text{M1}$$

$$\text{Answer: } 3+\sqrt{2}, \text{ so } a=3, b=1. \quad \text{A1}$$

(b) Given  $(2+\sqrt{3})x = 5+4\sqrt{3}$ , find  $x$  in the form  $a+b\sqrt{3}$ . [2]

$$x = \frac{5+4\sqrt{3}}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}}, \text{ denominator} = 4-3=1 \quad \text{M1}$$

$$\text{Numerator} = 10 - 5\sqrt{3} + 8\sqrt{3} - 12 = -2 + 3\sqrt{3}$$

$$\text{Answer: } x = -2 + 3\sqrt{3}, \text{ so } a = -2, b = 3. \quad \text{A1}$$

**Note:** In (b) the denominator rationalises to exactly 1 — a deliberate check that candidates still multiply out rather than guess.

**Question 3** — Compound angles**[4]**

$\sin A = \frac{-1}{\sqrt{5}}$  and  $\tan B = \frac{3}{4}$ , where  $A$  and  $B$  are in the same quadrant. Find  $p$  if  $\cos(A - B) = p\sqrt{5}$ .

$\sin A < 0 \Rightarrow A$  in quadrant 3 or 4;  $\tan B > 0 \Rightarrow B$  in quadrant 1 or 3. Same quadrant  $\Rightarrow$  both in quadrant 3. **B1**

Quadrant 3:  $\cos A = \frac{-2}{\sqrt{5}}$  (from  $\cos^2 A = 1 - \frac{1}{5}$ )

$\tan B = \frac{3}{4}$  in quadrant 3 gives  $\sin B = \frac{-3}{5}$ ,  $\cos B = \frac{-4}{5}$  (3–4–5 triangle, both negative) **B1**

$\cos(A - B) = \cos A \cos B + \sin A \sin B = \left(-\frac{2}{\sqrt{5}}\right)\left(-\frac{4}{5}\right) + \left(-\frac{1}{\sqrt{5}}\right)\left(-\frac{3}{5}\right)$  **M1**

$$= \frac{8}{5\sqrt{5}} + \frac{3}{5\sqrt{5}} = \frac{11}{5\sqrt{5}} = \frac{11\sqrt{5}}{25}$$

**Answer:**  $p = \frac{11}{25}$  **A1**

**Note:** Identifying quadrant 3 is the whole question — both signs negative. A candidate who takes quadrant 1 gets  $p = \frac{11}{25}$  by luck of sign pairing but cannot earn the two B marks.

**Question 4** — Completing the square and the discriminant**[4]**

(i) Express  $y = -2x^2 - 12x + 5$  as  $y = -a(x + b)^2 + c$ . [2]

$$-2x^2 - 12x + 5 = -2(x^2 + 6x) + 5 = -2[(x + 3)^2 - 9] + 5$$
 **M1**

**Answer:**  $y = -2(x + 3)^2 + 23$ , so  $a = 2$ ,  $b = 3$ ,  $c = 23$ . **A1**

(ii) Range of  $k$  for which  $-2x^2 - 12x + 5 = k$  has no two distinct roots. [2]

$$-2(x + 3)^2 + 23 = k \Rightarrow (x + 3)^2 = \frac{23 - k}{2}$$

Two distinct roots need  $\frac{23 - k}{2} > 0$ ; so NO two distinct roots when  $\frac{23 - k}{2} \leq 0$  **M1**

**Answer:**  $k \geq 23$  **A1**

**Note:** Equivalent by discriminant:  $2x^2 + 12x + (k - 5) = 0$ ,  $b^2 - 4ac = 144 - 8k \leq 0$ . Accept either route.  $k = 23$  MUST be included (equal roots are not two distinct roots) — a strict inequality loses the A1.

**Question 5 — Discriminant — line meets curve****[5]**

Show that  $y = x - 1$  intersects  $y = nx^2 + 3x - n$  at two distinct points for all real  $n$ .

**Equate:**  $x - 1 = nx^2 + 3x - n$  **M1**

$nx^2 + 2x + (1 - n) = 0$  **A1**

**Discriminant**  $= 2^2 - 4(n)(1 - n) = 4 - 4n + 4n^2 = 4(n^2 - n + 1)$  **M1**

$n^2 - n + 1 = \left(n - \frac{1}{2}\right)^2 + \frac{3}{4}$  **M1**

$\left(n - \frac{1}{2}\right)^2 \geq 0$ , so  $n^2 - n + 1 \geq \frac{3}{4} > 0$ , hence **discriminant**  $> 0$  for all real  $n$  — two distinct points.

**Shown.** **A1**

**Note:** The final A1 needs the explicit statement that the discriminant is positive — not merely that it is non-zero. Completing the square (or showing the discriminant of  $n^2 - n + 1$  is  $-3 < 0$ ) is required; asserting positivity without justification scores M1 only.

**Note:** Strictly the result needs  $n \neq 0$ , since  $n = 0$  makes the curve a straight line meeting  $y = x - 1$  once. The question's wording ("the curve") presumes  $n \neq 0$ ; do not penalise candidates either way.

**Question 6 — Logarithms — change of base****[5]**

Solve  $\log_2(x - 2) + 2\log_4(x - 3) = \frac{1}{2}\log_3 9$ .

**RHS:**  $\log_3 9 = 2$ , so  $\frac{1}{2}\log_3 9 = 1$  **B1**

$2\log_4(x - 3) = 2 \cdot \frac{\log_2(x - 3)}{\log_2 4} = 2 \cdot \frac{\log_2(x - 3)}{2} = \log_2(x - 3)$  **M1**

$\log_2(x - 2) + \log_2(x - 3) = 1 \Rightarrow \log_2[(x - 2)(x - 3)] = 1$

$(x - 2)(x - 3) = 2 \Rightarrow x^2 - 5x + 4 = 0 \Rightarrow (x - 1)(x - 4) = 0$  **M1**

$x = 1$  or  $x = 4$  **A1**

**Domain:** need  $x - 2 > 0$  AND  $x - 3 > 0$ , i.e.  $x > 3$ . **Reject**  $x = 1$ . **A1**

**Answer:**  $x = 4$

**Note:** The final A1 is for the rejection — it must be justified by the domain, not simply omitted. A candidate giving both roots scores a maximum of 4 out of 5.

- (i) Explain why the form  $\frac{A}{x+1} + \frac{B}{x^2+4}$  cannot be used. [2]

$x^2+4$  is an irreducible quadratic factor (it has no real linear factors, since  $x^2 = -4$  has no real solution). **B1**

A factor of degree 2 therefore requires a numerator of degree 1, i.e.  $Bx+C$ , not a constant. With only  $A$  and  $B$  there are two unknowns but three coefficients ( $x^2$ ,  $x$ , constant) to match, so no such  $A, B$  exist. **B1**

- (ii) Express  $\frac{3x^2+3x+10}{(x+1)(x^2+4)}$  in partial fractions. [4]

Let  $\frac{3x^2+3x+10}{(x+1)(x^2+4)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+4}$  **M1**

$$3x^2+3x+10 = A(x^2+4) + (Bx+C)(x+1)$$

$$x = -1: 3 - 3 + 10 = A(1+4) \Rightarrow 10 = 5A \Rightarrow A = 2$$
 **M1**

$$\text{Compare } x^2: 3 = A + B \Rightarrow B = 1$$
 **A1**

$$\text{Compare constants: } 10 = 4A + C = 8 + C \Rightarrow C = 2$$
 **A1**

**Answer:**  $\frac{2}{x+1} + \frac{x+2}{x^2+4}$

**Note:** Check by recombining:  $2(x^2+4) + (x+2)(x+1) = 2x^2+8+x^2+3x+2 = 3x^2+3x+10$ . ✓

Candidates who write  $\frac{B}{x^2+4}$  despite part (i) can score at most the first M1 in (ii).

$A(2,3), B(-1,-1), C(3,1)$ .

(i) Show  $AC \perp BC$ . [3]

$$m_{AC} = \frac{1-3}{3-2} = -2 \quad \text{M1}$$

$$m_{BC} = \frac{1-(-1)}{3-(-1)} = \frac{2}{4} = \frac{1}{2} \quad \text{M1}$$

$$m_{AC} \times m_{BC} = -2 \times \frac{1}{2} = -1, \text{ therefore } AC \perp BC. \text{ Shown.} \quad \text{A1}$$

(ii)  $ABCD$  is a parallelogram; find  $D$ . [2]

Diagonals of a parallelogram bisect each other, so midpoint  $AC$  = midpoint  $BD$ . M1

$$\text{Midpoint } AC = \left( \frac{2+3}{2}, \frac{3+1}{2} \right) = \left( \frac{5}{2}, 2 \right)$$

$$\frac{-1+x_D}{2} = \frac{5}{2} \Rightarrow x_D = 6; \quad \frac{-1+y_D}{2} = 2 \Rightarrow y_D = 5$$

**Answer:**  $D(6,5)$  A1

(iii) Find the area of  $ABCD$ . [2]

Right angle at  $C$ , so area =  $2 \times$  area of triangle  $ABC = 2 \times \frac{1}{2}(AC)(BC) = (AC)(BC)$  M1

$$AC = \sqrt{1^2 + (-2)^2} = \sqrt{5}, \quad BC = \sqrt{4^2 + 2^2} = 2\sqrt{5}$$

**Answer:** Area =  $\sqrt{5} \times 2\sqrt{5} = 10$  units<sup>2</sup> A1

**Note:** The shoelace formula on  $A(2,3), B(-1,-1), C(3,1), D(6,5)$  gives 10 as well — accept either method. Using the perpendicularity from (i) is the intended, shorter route.

$y = a \cos 2x + b$ ,  $a > 0$ , has maximum 4 and minimum  $-10$  for  $0 \leq x \leq 2\pi$ .

(i) Find  $a$  and  $b$ . [3]

Maximum:  $a + b = 4$ ; minimum:  $-a + b = -10$

**M1**

Adding:  $2b = -6 \Rightarrow b = -3$

**A1**

Then  $a = 4 - b = 7$

**A1**

(ii) State the period and the amplitude. [2]

Period  $= \frac{2\pi}{2} = \pi$

**B1**

Amplitude  $= 7$

**B1**

**Note:** Amplitude is  $a$ , \textit{not} the distance from maximum to minimum (14) and not affected by the shift  $b$ . This is the most common error on this question.

(iii) Sketch  $y = 7 \cos 2x - 3$  for  $0 \leq x \leq 2\pi$ . [2]

Cosine shape, 2 complete cycles over  $0 \leq x \leq 2\pi$

**B1**

Maximum 4 at  $x = 0, \pi, 2\pi$ ; minimum  $-10$  at  $x = \frac{\pi}{2}, \frac{3\pi}{2}$ ; oscillating about  $y = -3$

**B1**

**Note:** Curve must start at a maximum (cosine, positive  $a$ ) and the two cycles must be visibly equal in width.

Crossings of  $y = -3$  occur at  $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$ .

$P = 45\,000 e^{kt}$ ,  $t$  years after the beginning of 2015. At the beginning of 2016 the vase was valued at \$50 000.

(a) Show that  $k = 0.10536$  (5 s.f.). [3]

$$t = 1, P = 50\,000: 50\,000 = 45\,000 e^k \quad \text{M1}$$

$$e^k = \frac{50\,000}{45\,000} = \frac{10}{9} \quad \text{M1}$$

$$k = \ln \frac{10}{9} = 0.1053605\dots = 0.10536 \text{ (5 s.f.)}. \text{ Shown.} \quad \text{A1}$$

(b) Find the year in which the investment triples. [4]

$$\text{Tripled: } P = 3 \times 45\,000 = 135\,000 \quad \text{M1}$$

$$135\,000 = 45\,000 e^{kt} \Rightarrow e^{kt} = 3 \quad \text{M1}$$

$$t = \frac{\ln 3}{0.1053605} = \frac{1.098612}{0.1053605} = 10.427\dots \quad \text{M1}$$

$$t \approx 10.43 \text{ years after the beginning of 2015}$$

Answer: During 2025 A1

**Note:**  $t = 10$  is the beginning of 2025 and  $t = 11$  the beginning of 2026, so  $t = 10.43$  falls inside 2025. The A1 is for the YEAR, not for  $t$  — a candidate who stops at 10.4 years scores M1 M1 M1 only.

**Note:** Use the exact  $k = \ln \frac{10}{9}$  where possible; using the rounded 0.10536 is accepted here because the question supplied it.

(a) Term independent of  $x$  in  $3x^3(2x - \frac{1}{4x^2})^{12}$ . [3]

General term of  $(2x - \frac{1}{4x^2})^{12}$ :  $\binom{12}{r}(2x)^{12-r}(-\frac{1}{4x^2})^r$  M1

$$= \binom{12}{r} 2^{12-r} (-1)^r 4^{-r} x^{12-3r}$$

With the factor  $3x^3$  the power of  $x$  is  $3+12-3r=15-3r$ ; set  $15-3r=0 \Rightarrow r=5$  M1

$$\text{Term} = 3 \binom{12}{5} 2^7 (-1)^5 4^{-5} = 3 \times 792 \times 128 \times \frac{-1}{1024} = -3 \times 99$$

Answer:  $-297$  A1

**Note:**  $\binom{12}{5} = 792$  and  $\frac{128}{1024} = \frac{1}{8}$ . The sign comes from  $(-1)^5$  — losing it gives  $+297$  and costs the A1.

(b) In  $(2+mx)^{15}$  the ratio of the coefficient of  $x^7$  to that of  $x^9$  is  $\frac{4}{7}$ . Find  $m$ . [5]

Coefficient of  $x^7$ :  $\binom{15}{7} 2^8 m^7$  M1

Coefficient of  $x^9$ :  $\binom{15}{9} 2^6 m^9$  M1

$$\frac{\binom{15}{7} 2^8 m^7}{\binom{15}{9} 2^6 m^9} = \frac{4}{7}$$
 M1

$$\frac{6435}{5005} = \frac{9}{7}, \text{ so } \frac{9}{7} \cdot \frac{4}{m^2} = \frac{4}{7} \Rightarrow \frac{36}{7m^2} = \frac{4}{7}$$
 M1

$$m^2 = 9$$

Answer:  $m=3$  or  $m=-3$  A1

**Note:** Both values are required — the ratio involves  $m^2$ , so no root can be rejected. Giving only  $m=3$  scores 4 out of 5.

(a)  $f(x) = x^3 + x^2 + ax + 2b$  has factor  $(x+3)$ ;  $g(x) = x^3 - 4x^2 - ax - b$  leaves remainder 42 on division by  $(x-5)$ . Find  $a$  and  $b$ . [5]

$$(x+3) \text{ a factor} \Rightarrow f(-3) = 0: -27 + 9 - 3a + 2b = 0$$

M1

$$2b = 3a + 18 \dots (1)$$

A1

$$g(5) = 42: 125 - 100 - 5a - b = 42$$

M1

$$b = -5a - 17 \dots (2)$$

A1

$$\text{Substituting (2) into (1): } 2(-5a - 17) = 3a + 18 \Rightarrow -13a = 52$$

$$\text{Answer: } a = -4, b = 3$$

A1

**Note:** Check: (1) gives  $2(3) = 6$  and  $3(-4) + 18 = 6$  ✓. Sign slips in  $f(-3)$  ( $(-3)^3 = -27$ ,  $(-3)^2 = +9$ ) are the usual cause of lost A marks.

(b) Solve  $x^3 - x^2 - 7x + 7 = 0$ , giving the roots in exact form. [5]

$$\text{Group: } x^2(x-1) - 7(x-1) = 0$$

M1

$$(x-1)(x^2 - 7) = 0$$

A1

$$x^2 - 7 = 0 \Rightarrow x^2 = 7$$

M1

$$x = 1$$

A1

$$\text{Answer: } x = 1, x = \sqrt{7}, x = -\sqrt{7}$$

A1

**Note:** "Exact form" means  $\pm\sqrt{7}$  must be left in surd form — giving  $\pm 2.65$  loses the final A1. Factor theorem with trial  $x = 1$  is equally acceptable in place of grouping.

$L = A d^k$ , with  $d$ : 9, 50, 200, 400, 900 and  $L$ : 8, 15, 25, 32, 43.

(a) Explain how a straight line graph can be drawn, and draw it. [4]

Take  $\lg$  of both sides:  $\lg L = \lg A + k \lg d$

**B1**

Comparing with  $Y = mX + c$ : plot  $\lg L$  (vertical) against  $\lg d$  (horizontal); the graph is a straight line of gradient  $k$  and vertical intercept  $\lg A$ .

**B1**

$\lg d$ : 0.954, 1.699, 2.301, 2.602, 2.954

**M1**

$\lg L$ : 0.903, 1.176, 1.398, 1.505, 1.633

**A1**

(b) Estimate  $A$  and  $k$ . [4]

$$\text{Gradient } k = \frac{1.633 - 0.903}{2.954 - 0.954} = \frac{0.730}{2.000}$$

**M1**

**Answer:**  $k \approx 0.365$

**A1**

**Intercept:**  $\lg A = \lg L - k \lg d = 0.903 - 0.365(0.954) = 0.555$

**M1**

**Answer:**  $A = 10^{0.555} \approx 3.59$

**A1**

**Note:** These are graphical estimates: accept  $k$  in 0.35 to 0.38 and  $A$  in 3.3 to 3.9, provided the gradient is computed from two points on the DRAWN LINE and not from two raw data points.

**Note:** Check:  $3.59 \times 200^{0.365} = 3.59 \times 6.92 \approx 24.8$ , against the tabulated  $L = 25$  ✓

(a) Prove  $\frac{\cos \theta}{1 - \sin \theta} - \frac{1 - \sin \theta}{\cos \theta} = 2 \tan \theta$ . [5]

$$\text{LHS} = \frac{\cos^2 \theta - (1 - \sin \theta)^2}{(1 - \sin \theta) \cos \theta} \quad \text{M1}$$

$$(1 - \sin \theta)^2 = 1 - 2 \sin \theta + \sin^2 \theta \quad \text{M1}$$

$$\text{Numerator} = \cos^2 \theta - 1 + 2 \sin \theta - \sin^2 \theta; \text{ using } \cos^2 \theta = 1 - \sin^2 \theta: \quad \text{M1}$$

$$= 1 - \sin^2 \theta - 1 + 2 \sin \theta - \sin^2 \theta = 2 \sin \theta - 2 \sin^2 \theta = 2 \sin \theta (1 - \sin \theta) \quad \text{M1}$$

$$\text{LHS} = \frac{2 \sin \theta (1 - \sin \theta)}{(1 - \sin \theta) \cos \theta} = \frac{2 \sin \theta}{\cos \theta} = 2 \tan \theta = \text{RHS. Proved.} \quad \text{A1}$$

**Note:** Work on one side only. Candidates who cross-multiply from  $\text{LHS} = \text{RHS}$  have assumed the result and score a maximum of 3 (the M marks for the algebra).

(b) Hence solve  $\frac{\cos \theta}{1 - \sin \theta} - \frac{1 - \sin \theta}{\cos \theta} = \cot \theta$  for  $0 \leq \theta \leq 2\pi$ . [4]

$$\text{By (a): } 2 \tan \theta = \cot \theta = \frac{1}{\tan \theta} \quad \text{M1}$$

$$2 \tan^2 \theta = 1 \Rightarrow \tan \theta = \pm \frac{1}{\sqrt{2}} \quad \text{M1}$$

$$\text{Basic angle } \alpha = \tan^{-1} \frac{1}{\sqrt{2}} = 0.6155 \text{ rad}$$

$$\tan \theta = \frac{+1}{\sqrt{2}}: \theta = 0.6155, \pi + 0.6155 = 3.757 \quad \text{A1}$$

$$\tan \theta = \frac{-1}{\sqrt{2}}: \theta = \pi - 0.6155 = 2.526, 2\pi - 0.6155 = 5.668$$

$$\text{Answer: } \theta = 0.615, 2.53, 3.76, 5.67 \text{ rad (3 s.f.)} \quad \text{A1}$$

**Note:** All four solutions are required for the final A1. The range is in radians — answers in degrees score a maximum of 3. Note  $\theta \neq \frac{\pi}{2}, \frac{3\pi}{2}$  (undefined) but none of the solutions violate this.

(c) Find the set of values of  $k$  for which  $\frac{\cos \theta}{1 - \sin \theta} - \frac{1 - \sin \theta}{\cos \theta} = k(1 + \tan^2 \theta)$  has no solutions. [3]

$$\text{By (a) the equation becomes } 2 \tan \theta = k(1 + \tan^2 \theta). \text{ Let } t = \tan \theta: \quad \text{M1}$$

$$k t^2 - 2t + k = 0$$

$$\text{No real } t \text{ requires discriminant } < 0: (-2)^2 - 4(k)(k) < 0 \quad \text{M1}$$

$$4 - 4k^2 < 0 \Rightarrow k^2 > 1$$

$$\text{Answer: } k < -1 \text{ or } k > 1 \text{ (equivalently } k^2 > 1) \quad \text{A1}$$

**Note:**  $k = 0$  gives  $-2t = 0$ , i.e.  $t = 0$ , which DOES have solutions ( $\theta = 0, \pi, 2\pi$ ) — consistent, since  $0^2 < 1$ . Candidates who divide by  $k$  without considering  $k = 0$  should not be penalised provided the final set is correct.